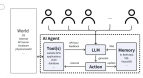
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Overview of LLM-based AI agent.

Source publication

Security of AI Agents

Preprint | Full-text available

Jun 2024

Yifeng He · Ethan Wang · Yuyang Rong · [...] · Hao Chen

The study and development of AI agents have been boosted by large language models. AI agents can function as intelligent assistants and complete tasks on behalf of their users with access to tools and the ability to execute commands in their environments, Through studying and experiencing the workflow of typical AI agents, we have raised several co...

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Contexts in source publication

Context 1

... AI agents depend on their AI model to understand user input and environment feedback and generate actions to use tools, we say that the AI model is the backbone of the agent. We summarize the basic architecture of LLM-based AI agents in Figure 1 . Traditional agents operate on pre-defined rules [1,2] or reinforcement learning models [3], making them hard to generalize to new tasks and different tools. ...

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Context 2

... keeps the results of the model's weight updates isolated per user. The challenge of this method lies in the significant costs associated with fine-tuning Figure 1 0: Session-aware AI agents with prompt tuning. I µI¼□ I µIz□I µIz□ denotes the added trainable parameters only for the user's chat history. ...

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Context 3

... µI¼□ I µIz□ are designed to only remember past prompts or chat history. As described in Figure 1 0, prompt tuning is particularly well-suited for AI agents as it allows each agent to fine-tune individual instances of I µI¼□ I µIz□ for each user. This method ensures that no chat history is shared with the foundation model provider, thereby addressing potential privacy concerns discussed in subsection 3.2 at their root. ...

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Context 4

... other method for advancing AI agents involves enhancing their learning capabilities through memory updates. The CoALA architecture [19] formulates the Figure 1 1: Session-aware AI agents with updatable episodic memory. When learning only by only updating episodic memory, the AI agent can improve itself without polluting the foundation LLM. ...

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Context 5

... practice, agents maintain these memories in a vector database, which then feeds relevant information to the LLM through retrievalaugmented generation (RAG) [70]. This method allows the agents to enhance their performance or become more personalized by exclusively updating their episodic and semantic memories, as shown in Figure 1 1. Although such updates do not directly enhance the LLM's intrinsic intelligence, they can significantly increase an agent's applicability and effectiveness in assisting users. ...

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... HITL enables explainability, auditability, and trust, which are key requirements when AI-generated outputs may affect technical or financial decisions. Recent studies emphasized that in many industrial domains, fully autonomous AI agents may not be justified, and AI is more effective when deployed as a supervised contributor, particularly in contexts demanding trust, reliability, and regulatory compliance (He et al., 2024; Kotsiopoulos et al., 2024). ...

Potential of Large Language Models for Construction Quotation Workflows

Conference Paper

Jul 2026

Anja Brelih · Žiga Krvina · Robert Klinc

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... Autonomous AI agents are software entities capable of perceiving complex environments, reasoning across multi-step tasks, and executing actions without direct human instruction, with persistent access to communication channels, external APIs, databases, and public publishing platforms (Gao & Wu, 2025;Gürpinar, 2025). Powered by large language models, these agents integrate memory systems, tool usage, and long-horizon planning mechanisms, shifting AI from reactive tools to proactive operational entities in cybersecurity, financial services, and AI governance (Yigit et al., 2025; He et al., 2024) . Unrestricted communication capabilities allow agents to interact autonomously with social media platforms and organizational databases, amplifying their utility. ...

Security Risks of Autonomous AI Agents with Unrestricted Communication and Publishing Capabilities

Article | Full-text available

May 2026

Oluwadayo Mafolasere Olaniyi · Suleiman S. Abba · Christopher Ugbong Akeke · Faith Hauwa Oluwapamilerin Kolo · Moses Abuobelye Akeke

Autonomous AI agents powered by large language models increasingly operate with unrestricted communication and publishing capabilities, creating critical security vulnerabilities through memory poisoning attacks that corrupt long-term agent memory and propagate falsified outputs to public...

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... 8 This scaffolding allows a foundation model to interact with various tools and services, enabling it to execute plans and take actions in the world. He et al. 2024) We highlight four core components in foundation model-based agents that have been previously identified in the literature (Weng 2023;L. Wang et al. 2024). ...

AI Agent Governance: A Field Guide

Preprint | Full-text available

May 2025

Jam Kraprayoon · Zoe Williams · Rida Fayyaz

This report serves as an accessible guide to the emerging field of AI agent governance. Agents - AI systems that can autonomously achieve goals in the world, with little to no explicit human instruction about how to do so - are a major focus of leading tech companies, AI start-ups, and investors. If thes...

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... Because the governance of AI agents remains a nascent topic, potential liability issues in the rapidly expanding agentic market are prevalent but not sufficiently analyzed. Existing efforts are built along two streams: one focuses on establishing and refining the taxonomies of risks and harms using empirical evidence (Chan et al., 2023(Chan et al., , 2024 He et al., 2024; Hammond et al., 2025) or differentiating the governance of agentic systems from traditional machine learning (ML) models Kolt, 2025); another focuses on under-standing the interactions between humans and AI agents to build constructive principles (Zheng et al., 2023). Research in both streams used principalagent theory (PAT) (Eisenhardt, 1989;Laffont and Martimort, 2002) as a starting point (see Appendix C), but lacked a systematic examination of how existing legal frameworks for liability can effectively address various principal-agent relationships in AI systems. ...

Inherent and emergent liability issues in LLM-based agentic systems: a principal-agent perspective

Conference Paper

Jan 2025

Garry A. Gabison · R. Patrick Xian

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